

Bitter Pills to Swallow: The Enforcement Costs of Health Litigation

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Abstract

Public health procurement is shaped not only by administrative decisions but also by judicial interventions that compel urgent purchases. These court-mandated procurements carry substantial fiscal costs because they undermine procurement planning and expose officials to sanctions that incentivize compliance over cost-efficiency—a mechanism we term the “under the gun” effect. Using granular bid-level data on pharmaceutical procurement in São Paulo, Brazil (2009–2019), with item, time, and buyer fixed effects, we estimate that judicial enforcement increases reference prices by 2.7–17.8%, raises negotiated prices by 3.1–17.2%, and reduces bidder participation by 4.1–10.7%. Comparing litigated purchases directly to ordinary ones—excluding administrative purchases—yields even larger effects: reference prices rise by 7.6%, negotiated prices by 9.1%, and bidder participation falls by 7.9%. Isolating the sanction channel by comparing litigated and administrative urgent purchases, we find judicial pressure alone raises prices by 23.2–30.2%, driven by 27% higher reference price ceilings and substantially smaller order quantities. These findings highlight the economic burden of judicial enforcement on public budgets and suggest that legal mandates should weigh procurement efficiency alongside individual rights.

Keywords: public procurement, health litigation, enforcement costs, judicial mandates, Brazil

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1. Introduction

When courts order governments to purchase specific goods, they override standard procurement procedures and compress the time available for planning and competitive bidding. This paper estimates the fiscal costs of such judicial enforcement in the context of health litigation in Brazil, where courts routinely compel state agencies to procure prescription drugs under tight deadlines.

The interaction between the judiciary and public administration has long attracted scholarly attention. Courts and regulatory agencies can be viewed as alternative instruments of social control, each with distinct comparative advantages ([Glaeser and Shleifer, 2003](#)). While courts protect individual rights, their interventions in procurement can be costly: inefficient judicial processes may induce public buyers to accept suboptimal outcomes ([Coviello et al., 2018](#)). In health care, judicial mandates represent a particularly consequential form of intervention, as they directly determine how—and at what price—the government acquires essential medicines ([Wang, 2015](#); [Ferraz, 2009](#)).

Brazil provides an ideal setting to study these costs. The 1988 Constitution enshrines health as a “right of all and a duty of the state,” and courts have adopted an expansive interpretation under which virtually any medication can be obtained through litigation, regardless of cost or inclusion on the formulary of the Unified Health System (SUS). Court orders are enforced as preliminary injunctions requiring purchase within days, compared to the weeks or months available for ordinary procurement. The volume of health litigation has grown dramatically: first-instance health claims reached approximately 96,000 nationally in 2017, a 130% increase over 2008 ([CNJ/INSPER, 2019](#)). In São Paulo state alone, health lawsuits grew by 913% between 2008 and 2017.

We construct a unique bid-level dataset covering all pharmaceutical procurement transactions by the São Paulo State Department of Health (SES/SP) from January 2009 through December 2019, conducted through the state’s electronic procurement platform (BEC). The dataset comprises over 479,000 purchase-offer-item (POI) observations for medical supplies (BEC Group 65), encompassing ordinary, administrative, and litigated purchases. We classify purchase types using a regular expression algorithm applied to

public tender notices, cross-referenced with administrative records from the S-CODES health litigation database.

Our empirical strategy exploits the quasi-experimental variation in procurement conditions generated by court orders and administrative requests. These exogenous shocks partition purchases into two regimes: *ordinary* (standard planning timeline) and *urgent* (compressed timeline due to judicial or administrative mandate). Within urgent purchases, we further distinguish *administrative* requests—which carry no penalty for procurement failure—from *litigated* purchases, where officials face severe sanctions including fines, personal liability, and fund seizure. This distinction allows us to isolate what we term the “under the gun” effect: the additional cost attributable to the threat of judicial punishment.

We estimate high-dimensional fixed effects regressions with item, year (or year-month), and public buyer unit (PBU) fixed effects, clustering standard errors at the PBU level. Our preferred specification (item + year + PBU FE) compares the same product purchased by the same buyer across ordinary and urgent procurement episodes, absorbing time-invariant item heterogeneity, common time trends, and buyer-specific factors.

The main findings are as follows. First, urgent purchases carry higher reference prices—the maximum price the government is willing to pay—by 2.7% in our preferred specification. Second, negotiated prices are 5.4% higher for urgent purchases (total effect), with a direct effect of 3.1% after controlling for quantity differences. Third, urgent tenders attract 5.5% fewer bidding firms. Fourth, the probability of tender success is 2.1 percentage points *higher* for urgent purchases, consistent with procurement officials accepting higher prices to ensure compliance rather than risk tender failure.

We further isolate the effect of court orders by comparing litigated purchases directly to ordinary ones, excluding administrative purchases. This yields larger estimates across all outcomes: reference prices are 7.6% higher, negotiated prices 9.1% higher, and bidder participation 7.9% lower for litigated purchases. The entire negotiated price premium operates through the quantity channel: controlling for quantity, the direct effect is economically zero.

The “under the gun” analysis reveals that the sanction channel is quantitatively important. Comparing litigated and administrative urgent purchases for the same items, we find that litigated tenders carry prices 23.2–30.2% higher in our preferred specifications, a difference attributable to the threat of judicial punishment. Extending this analysis to all outcomes, we show that the sanction channel operates primarily through reference prices—which are 27% higher for litigated than administrative purchases—and through quantities, with litigated purchases involving substantially smaller orders that forgo bulk discounts.

Our heterogeneity analysis shows that the enforcement costs vary across several dimensions. The price premium of urgent purchases is concentrated among items in the specialized SUS component (non-basic medicines), in the later period (2014–2019), and in markets with lower competition. These patterns are consistent with a model in which judicial pressure interacts with market structure to amplify procurement inefficiency.

This paper makes three contributions to the literature. First, we provide the first estimates of the causal effect of judicial enforcement on public procurement outcomes using comprehensive administrative data and high-dimensional fixed effects, contributing to the literature on government spending efficiency ([Bandiera et al., 2009](#); [Best et al., 2023](#); [Decarolis et al., 2020](#)). Second, we identify and quantify the “under the gun” effect—the cost premium attributable to the threat of judicial sanctions—by exploiting the institutional distinction between administrative and litigated urgent purchases. This contributes to the growing literature on how oversight and accountability mechanisms can generate unintended costs ([Rasul and Rogger, 2018](#); [Williams, 2021](#); [Duflo et al., 2012](#)). Third, we document how judicial mandates distort the planning phase of procurement, adding to the evidence on the importance of ex ante design in auction performance ([Coviello et al., 2018](#); [Szücs, 2023](#); [Baltrunaite et al., 2021](#)).

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the institutional background. Section 4 presents the data and sample. Section 5 details the empirical strategy. Section 6 presents the results. Section 7 concludes.

2. Related Literature

This paper contributes to several strands of the economics literature.

Government spending efficiency and procurement. A foundational distinction in the study of public spending separates *active waste* (corruption) from *passive waste* (inefficiency), as formalized by [Bandiera et al. \(2009\)](#), who estimate that passive waste accounts for the majority of excess spending in Italian public procurement. Subsequent work has identified specific channels of passive waste, including bureaucratic capacity ([Best et al., 2023](#)), management practices ([Rasul and Rogger, 2018](#); [Bloom et al., 2015](#)), and the design of procurement mechanisms ([Lewis-Faupel et al., 2016](#); [Szücs, 2023](#)). Our paper identifies a previously unstudied source of passive waste: the compression of planning time induced by judicial mandates. We show that this channel operates through both the reference price setting process (the internal phase) and the competitive bidding process (the external phase).

Procurement design and competition. A growing literature documents how procurement design affects competition and prices. [Coviello et al. \(2018\)](#) show that court-imposed delays reduce bidder participation and increase procurement costs in Italian public works. [Decarolis et al. \(2020\)](#) demonstrate that buyer competence is a key determinant of procurement outcomes. [Baltrunaite et al. \(2021\)](#) find that reducing discretion in procurement can improve outcomes when corruption is prevalent but may harm outcomes otherwise. Our setting differs in that judicial intervention compresses rather than extends the procurement timeline, and the primary mechanism operates through planning disruption rather than direct manipulation of award criteria.

Accountability, oversight, and unintended consequences. A related literature examines how accountability mechanisms intended to reduce waste can themselves generate costs. [Rasul and Rogger \(2018\)](#) show that management practices in the Nigerian civil service affect project completion rates, while [Williams \(2021\)](#) finds that management quality explains substantial variation in public service delivery. [Duflo et al. \(2012\)](#) argue that the

design of incentives for public officials must balance accountability with operational flexibility. Our “under the gun” effect is a specific instance of this tension: judicial sanctions intended to ensure compliance with health mandates inadvertently raise procurement costs by distorting the bargaining environment.

Health litigation and right-to-health enforcement.. The legal and public health literatures have documented the growth of health litigation in developing countries, particularly in Latin America (Wang, 2015; Ferraz, 2009; Biehl et al., 2009). CNJ/INSPER (2019) provide comprehensive data on the judicialization of health in Brazil. However, the existing literature has focused primarily on the legal and public health dimensions, with limited attention to the fiscal costs transmitted through the procurement channel. Our paper fills this gap by providing granular estimates of how judicial enforcement affects specific procurement outcomes—prices, quantities, competition, and tender success.

Auction theory under urgency.. From a theoretical perspective, our findings connect to the auction theory literature on how the number of bidders and time pressure affect prices. Bulow and Klemperer (1996) show that attracting an additional bidder can be more valuable than optimizing the auction mechanism. When judicial mandates reduce bidder participation, they directly undermine the competitive benefits of procurement auctions. Moreover, the public information that a tender is court-mandated creates an asymmetric bargaining environment that further erodes the government’s position (Krishna, 2009).

3. Institutional Background

This section describes the institutional framework governing health litigation and pharmaceutical procurement in Brazil, with a focus on the state of São Paulo.

3.1. Health as a Constitutional Right and the Rise of Litigation

The 1988 Brazilian Constitution created the Unified Health System (SUS), a universal public health system jointly funded and administered by federal, state, and municipal governments. The SUS maintains a formulary of approved medications and procedures, periodically updated to reflect medical advances subject to cost-effectiveness criteria. This

formulary represents the society’s chosen balance between universal coverage and fiscal sustainability (Castro et al., 2019; Soares, 2019).

However, Brazilian courts have adopted a strict literal interpretation of Article 196 of the Constitution, which states that “health is a right of all and a duty of the state.” Under this interpretation, any individual can obtain virtually any medication through litigation, irrespective of cost or inclusion on the SUS formulary. The range extends from acetylsalicylic acid to galsulfase (for mucopolysaccharidosis type VI, with annual treatment costs of approximately US\$400,000).

Several factors make health litigation particularly prevalent. Access to the legal system is inexpensive—requiring only a prescription and a public defender—and success rates are exceptionally high: approximately 85% of first-instance claims in São Paulo are granted (CNJ/INSPER, 2019), with even higher rates on appeal. These conditions create strong incentives for individuals to seek medicines through the courts rather than through standard SUS channels.

The growth in health litigation has been dramatic. Nationally, first-instance health court orders reached approximately 96,000 in 2017, representing a 130% increase over 2008 (CNJ/INSPER, 2019). In São Paulo, the increase was 913%, from 2,317 to 23,465 annual lawsuits between 2008 and 2017. Court orders span a wide geographic distribution across the state’s municipalities (Figure 1) and encompass approximately 2,760 distinct items ordered at least once per year.

The fiscal impact is substantial. São Paulo’s annual public health budget was approximately US\$5.9 billion in 2018 (about 10% of the state’s total budget). Public spending on litigated health items alone amounted to nearly 5% of this budget, or approximately US\$300 million annually.

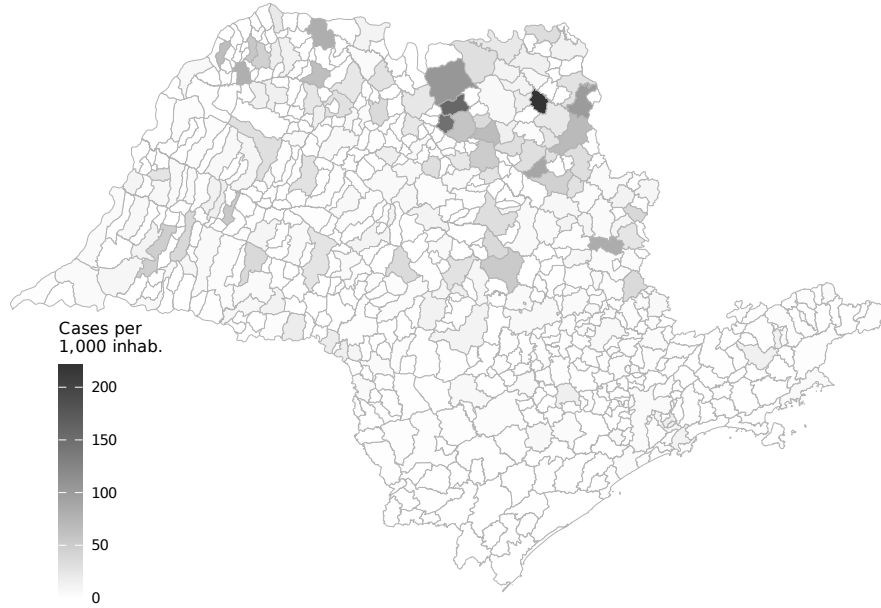


Figure 1: Health Litigation Cases per 1,000 Inhabitants Across Municipalities in São Paulo (2009–2019)

Source: Authors' elaboration based on CNJ/INSPER (2019) litigation data and IBGE population estimates (SIDRA table 6579, 2009–2019 average).

3.2. Public Procurement under Judicial Pressure

The São Paulo State Department of Health (SES/SP) manages health policy implementation through 99 decentralized public buyer units (PBUs) distributed across the state (Figure 2).

Federal Law 8,666/1993 establishes the general framework for public procurement, requiring that all government purchases follow a formal tendering process organized in three sequential phases:

1. **Internal phase (planning):** PBUs identify needs, specify items, set quantities and reference prices, choose the auction format, and publish a public notice.
2. **External phase (bidding):** Firms compete through a chosen competitive procedure—either sealed-bid (*convite*, up to R\$176,000) or multi-round descending auction (*pregão*, no value limit).
3. **Delivery phase:** The winning firm delivers the procured items.



Figure 2: Public Buyer Units (SES/SP)

Source: Authors' elaboration based on SES/SP administrative records.

For ordinary purchases, the internal phase typically spans 30 to 180 days, allowing PBUs to aggregate demand, research market prices, and set competitive reference prices. Court orders fundamentally disrupt this process. Almost all court decisions (99.94%) are enforced as preliminary injunctions with delivery deadlines of 1 to 10 days, compressing the internal phase to roughly one-third of the ordinary timeline.

This compression affects procurement in several ways. First, PBUs lose the ability to aggregate demand across units and over time, reducing their bargaining leverage from bulk purchasing. Second, reference prices must be set quickly with limited market research, often resulting in higher reservation prices. Third, approximately 75% of litigated items are not on the SUS formulary, meaning PBUs have less experience procuring them. Fourth, roughly 65% of court orders involve “packages” combining SUS and non-SUS items, further complicating procurement planning.

3.3. *The Sanction Channel: Administrative vs. Litigated Purchases*

A critical institutional feature for our identification strategy is the distinction between *litigated* and *administrative* urgent purchases. Both types face identical planning constraints—compressed timelines, small quantities, and diverted budget resources. However, they differ sharply in the consequences of procurement failure.

For litigated purchases, failure to comply with a court order exposes public officials to severe sanctions: substantial fines (sometimes disproportionate to the purchase value), administrative and criminal liability, and seizure or blocking of public funds. These penalties are public information, as the court order must be referenced in the tender notice.

Since 2009, the SES/SP has attempted to mitigate these costs through administrative requests, a mechanism allowing the government to negotiate supply directly with individuals before litigation occurs. Administrative requests undergo evaluation by a scientific committee using evidence-based medicine criteria. Crucially, failure to complete an administrative purchase carries no penalties for public officials. Between 2009 and 2019, administrative requests generated approximately 9,700 purchases, or about 6.5% of the total from court orders.

This institutional distinction provides a natural comparison group for isolating the “under the gun” effect—the additional procurement cost attributable to the threat of judicial sanctions—since administrative and litigated purchases share all planning and execution constraints but differ only in the penalty regime.

	Ordinary	Administrative	Litigated
Source of funds	Dedicated budget	Diverted budget	Diverted budget
Quantity	Large	Small	Small
Delivery time	Long	Short	Short
Threat of punishment	None	None	Potential punishment

Figure 3: Types of Purchases

Source: Authors' elaboration.

3.4. The Bidding Process

Both ordinary and urgent purchases are conducted through the *Bolsa Eletrônica de Compras* (BEC), São Paulo state's electronic procurement platform, mandatory for all common goods and services since 2007. The BEC handles substantial volumes: approximately US\$1.7 billion in SES/SP transactions in 2018, and over R\$7.4 billion since 2009.

Two competitive procedures are used. In sealed-bid tendering (*convite*), firms submit proposals by a deadline; the lowest qualifying bid wins. In multi-round descending auctions (*pregão*), an initial sealed-bid phase is followed by a 20-minute reverse auction where firms observe the current lowest bid and can submit lower bids. If a bid is placed in the final four minutes, the auction extends by four additional minutes. After the auction, a negotiation phase allows further price reductions.

The key implication for our analysis is that all purchase types use the same electronic platform and competitive procedures. The variation we exploit arises exclusively from differences in planning conditions and penalty regimes, not from differences in the auction mechanism itself.

4. Data and Sample

4.1. Data Sources

We combine three administrative datasets covering January 2009 through December 2019.

Public procurement data.. Our primary dataset consists of bid-level records from the BEC electronic procurement platform for all SES/SP purchases of medical supplies classified under BEC Group 65 (“Medical, dental, and hospital equipment and supplies”). Each observation corresponds to a purchase-offer-item (POI)—the combination of a purchase order and a specific item—and includes information on: (i) the internal phase (quantities demanded, reference prices, PBU identifier, auction format); (ii) the external phase (bid prices from all participants, number of bidding firms, tender outcome); and (iii) the winning bid (final negotiated price, firm identifier).

Purchase type classification.. We classify each POI as ordinary, administrative, or litigated using a regular expression (REGEX) algorithm applied to the “Object of the Contract” field of public tender notices. This text field describes the purchase purpose and, by law, must reference any underlying court order or administrative request.

Health litigation records.. Data on individual health claims come from S-CODES, the SES/SP administrative database of all health-related lawsuits against the state of São Paulo. We derive several variables from this source, including SUS formulary status, whether the claim involves a drug package, and whether the order was enforced through a preliminary injunction.

4.2. Sample Construction

The raw dataset comprises 479,330 POI observations across 32,532 distinct items, 94,182 purchase orders, and 100 PBUs. Table 1 reports descriptive statistics by purchase type.

The analysis sample is restricted to items for which at least one ordinary and at least one litigated purchase is observed during the sample period, ensuring within-item comparability across procurement regimes. This restriction yields a sample of approximately 226,000 observations covering 3,856 items. For regressions using bid prices and quantities, we further restrict to winning bids (approximately 197,000 observations).

Several features of the data are noteworthy. In levels, ordinary purchases have substantially higher mean reference prices (R\$909) compared to administrative (R\$226) and

Table 1: Descriptive Statistics by Purchase Type

	Ordinary		Administrative		Litigated		Diff	Diff
	Mean	(SD)	Mean	(SD)	Mean	(SD)	(O-L)	(A-L)
<i>Panel A: Levels</i>								
Reference Price	909.17	(5,959.41)	226.38	(2,257.35)	370.39	(2,540.20)	538.78*** [0.000]	-144.01*** [0.000]
Negotiated Price	477.51	(2,915.88)	148.03	(1,177.52)	260.25	(1,561.81)	217.26*** [0.000]	-112.22*** [0.000]
Quantity	31,487.15	(163,507.29)	117,146.00	(356,962.37)	9,646.19	(82,683.10)	21,840.96*** [0.000]	107,499.81*** [0.000]
N. Bidding Firms	3.17	(1.93)	2.43	(1.35)	2.20	(1.19)	0.97*** [0.000]	0.23*** [0.000]
<i>Panel B: Log Transformations</i>								
Log Reference Price	0.926	(2.723)	0.909	(2.541)	1.557	(2.446)	-0.632*** [0.000]	-0.648*** [0.000]
Log Negotiated Price	0.319	(2.862)	0.480	(2.604)	1.024	(2.568)	-0.705*** [0.000]	-0.544*** [0.000]
Log Quantity	6.947	(2.678)	7.472	(3.160)	6.087	(2.083)	0.860*** [0.000]	1.385*** [0.000]
Log N. Firms	0.984	(0.590)	0.747	(0.529)	0.665	(0.493)	0.319*** [0.000]	0.081*** [0.000]
<i>Panel C: Tender Characteristics</i>								
Successful Tender (%)	0.859	(0.348)	0.869	(0.338)	0.908	(0.288)	-0.050*** [0.000]	-0.040*** [0.000]
Observations	136,250		15,371		45,351			

Notes: Sample restricted to items with at least one ordinary and one litigated purchase. Winners only for price and quantity variables. Variables winsorized at 1%/99%. Stars on differences from Welch *t*-tests; *p*-values in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

litigated (R\$370) purchases, reflecting the mix of items across procurement types. However, the composition effect is absorbed by our item fixed effects. In logs—which form the basis of our regressions—litigated purchases show higher mean prices (1.56 for reference price, 1.02 for negotiated price) than ordinary purchases (0.93 and 0.32, respectively), consistent with a price premium for urgent procurement conditional on the item.

Quantities are markedly lower for litigated purchases (mean of 9,646 units vs. 31,487 for ordinary), reflecting the inability to aggregate demand under compressed timelines. The number of bidding firms is also lower for urgent purchases (2.2 for litigated vs. 3.2 for ordinary), suggesting reduced competition in the external phase.

For the “under the gun” analysis, we construct a subsample of urgent purchases (administrative and litigated) for items with both types present, yielding approximately 57,000 observations. Table 2 presents balance statistics for this subsample.

Table 2: Balance Table: Administrative vs Litigated (Urgent Purchases)

	Administrative		Litigated		Diff	<i>p</i> -value
	Mean	(SD)	Mean	(SD)	(A-L)	
<i>Panel A: Procurement Outcomes</i>						
Reference Price	226.38	(2,257.35)	232.56	(1,432.18)	-6.18	[0.752]
Negotiated Price	148.03	(1,177.52)	174.79	(1,052.56)	-26.75**	[0.013]
Quantity	117,146.00	(356,962.37)	9,306.87	(80,252.25)	107,839.13***	[0.000]
Log Reference Price	0.91	(2.54)	1.43	(2.32)	-0.52***	[0.000]
Log Negotiated Price	0.48	(2.60)	0.89	(2.44)	-0.41***	[0.000]
Log Quantity	7.47	(3.16)	6.18	(1.99)	1.29***	[0.000]
<i>Panel B: Market Structure</i>						
N. Bidding Firms	2.427	(1.351)	2.155	(1.120)	0.272***	[0.000]
Log N. Firms	0.747	(0.529)	0.650	(0.483)	0.097***	[0.000]
Successful Tender (%)	0.869	(0.338)	0.914	(0.280)	-0.045***	[0.000]
<i>Panel C: Purchase Characteristics</i>						
Electronic Auction	0.985	(0.123)	0.941	(0.236)	0.044***	[0.000]
Observations	15,371		41,491			

Notes: Sample restricted to urgent purchases (administrative and litigated) for items with both types present. Winners only for price/quantity. Variables winsorized at 1%/99%. *p*-values from Welch *t*-tests in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

4.3. Variable Definitions

Our key variables are defined as follows:

- **Urgent:** indicator equal to 1 for administrative or litigated purchases (purchase type $\in \{1, 2\}$), 0 for ordinary purchases.
- **Administrative:** indicator equal to 1 for administrative purchases, 0 for litigated. Used in the “under the gun” subsample analysis.
- **Reference price:** the maximum price specified in the tender notice. We use the log transformation.
- **Negotiated price:** the final price of the winning bid. Log transformation.
- **Quantity:** the quantity demanded in the tender notice. Log transformation.
- **Number of firms:** the count of distinct firms submitting bids. Log transformation.

- **Tender success:** indicator equal to 1 if the tender resulted in a successful purchase.

All continuous variables are winsorized at the 1st and 99th percentiles to limit the influence of outliers. Robustness checks with no winsorization and 5%/95% winsorization are reported in the Appendix.

5. Empirical Strategy

Our empirical analysis proceeds in two stages. First, we estimate the overall enforcement costs of urgent procurement by comparing ordinary and urgent purchases. Second, we isolate the “under the gun” effect by comparing litigated and administrative urgent purchases.

5.1. Identification

The key identification assumption is that, conditional on item, time, and buyer fixed effects, the assignment of purchases to the urgent category is exogenous to unobserved determinants of procurement outcomes. Several institutional features support this assumption.

First, judicial orders and administrative requests originate from individuals’ health needs and legal claims, which are unrelated to the procurement process itself. The principle of impersonality in Brazilian public administration ensures that tender outcomes depend on purchase characteristics—item specifications, market conditions, and planning parameters—not on the identity or circumstances of the requesting individual.

Second, the timing and content of court orders are unpredictable from the PBU’s perspective. PBUs cannot anticipate exactly which items, in what quantities, and when they will be compelled to purchase. Only 4% of judicial claims arise from stock shortages or service failures, suggesting that litigation is poorly correlated with PBU-level mismanagement or unobserved quality.

Third, the key source of variation—the urgency regime—is determined by external judicial or administrative decisions, not by PBU choice. While PBUs might in principle manipulate item classification, the legal requirement to reference court orders in tender notices and our REGEX-based classification using official texts mitigate this concern.

5.2. *Enforcement Costs: Ordinary vs. Urgent*

We estimate the following baseline specification:

$$y_{i,g,t} = \beta \cdot \text{Urgent}_{i,g,t} + \gamma_g + \lambda_t + \delta_b + \varepsilon_{i,g,t}, \quad (1)$$

where $y_{i,g,t}$ is the outcome (log reference price, log negotiated price, log quantity, log number of firms, or tender success) for purchase order i of item g at time t ; $\text{Urgent}_{i,g,t}$ equals 1 for administrative or litigated purchases and 0 for ordinary purchases; γ_g are item fixed effects; λ_t are time fixed effects (year or year-month); and δ_b are PBU fixed effects. Standard errors are clustered at the PBU level.

We report four specifications with progressively richer fixed effects:

- (1) Item FE only;
- (2) Item + Year FE;
- (3) Item + Year + PBU FE (preferred);
- (4) Item + Year-Month + PBU FE.

Specification (3) is our preferred specification because it absorbs time-invariant item characteristics, common annual trends, and buyer-specific heterogeneity, while retaining sufficient within-cell variation for precise estimation.

For negotiated prices and number of firms, we additionally estimate a “direct effect” specification that controls for log quantity:

$$y_{i,g,t} = \beta \cdot \text{Urgent}_{i,g,t} + \alpha \cdot \ln Q_{i,g,t} + \gamma_g + \lambda_t + \delta_b + \varepsilon_{i,g,t}. \quad (2)$$

This specification captures the direct effect of urgency on prices and competition, net of the quantity channel (since urgent purchases typically involve smaller quantities, which may independently affect prices through reduced bulk discounts).

5.3. *The “Under the Gun” Effect: Litigated vs. Administrative*

To isolate the effect of judicial sanctions on procurement costs, we restrict the sample to urgent purchases only (both administrative and litigated) for items with at least one

purchase of each type, and estimate:

$$\ln(\text{Price}_{i,g,t}) = \beta \cdot \text{Admin}_{i,g,t} + \gamma_g + \lambda_t + \delta_b + \varepsilon_{i,g,t}, \quad (3)$$

where Admin equals 1 for administrative purchases and 0 for litigated purchases. The coefficient β captures the price differential between administrative and litigated purchases, holding constant the item, time period, and buying unit. Since both types face identical planning constraints, β isolates the cost attributable to the sanction regime: the threat of judicial punishment for procurement failure.

6. Results

6.1. Reference Prices

Table 3 presents estimates of the effect of urgent procurement on log reference prices—the maximum price the government is willing to pay, set during the internal planning phase.

Table 3: Reference Prices

	(1)	(2)	(3)	(4)
Urgent Purchase	0.164*** (0.056)	0.156*** (0.050)	0.027* (0.014)	0.027* (0.014)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	196,899	196,899	196,896	196,896
Within R ²	0.003	0.003	0.000	0.000

Notes: Dependent variable: log reference price. Sample: winners only, items with both ordinary and litigated purchases. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The coefficient on Urgent Purchase is positive and statistically significant across all specifications. In the most parsimonious specification with item FE only (column 1), urgent purchases carry reference prices that are 17.8% higher ($e^{0.164} - 1$). Adding year FE

(column 2) yields a similar estimate of 16.9%. The magnitude decreases substantially in our preferred specification with item + year + PBU FE (column 3), where the premium is 2.7% ($e^{0.027} - 1$, $p < 0.10$), and remains stable at 2.7% with year-month FE (column 4).

The attenuation from columns (1)–(2) to columns (3)–(4) is informative. It indicates that a substantial share of the raw price premium in urgent procurement is absorbed by PBU fixed effects, suggesting that certain buying units systematically handle more urgent purchases and also face higher prices for unrelated reasons. Our preferred estimate of 2.7% represents the within-item, within-buyer, within-year premium attributable to urgency.

6.2. Quantities

Table 4 reports estimates for log quantities demanded.

Table 4: Quantities

	(1)	(2)	(3)	(4)
Urgent Purchase	-0.264 (0.183)	-0.279 (0.181)	-0.058 (0.056)	-0.065 (0.058)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	196,899	196,899	196,896	196,896
Within R ²	0.003	0.003	0.000	0.000

Notes: Dependent variable: log quantity. Sample: winners only. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The point estimates are negative across all specifications, consistent with reduced quantities under urgency, but are not statistically significant at conventional levels in our preferred specifications (columns 3–4). The magnitude of -0.058 in column (3) corresponds to approximately 5.6% fewer units. The imprecision likely reflects the high variance in quantities across purchase orders and the fact that quantities are partially determined by the nature of the court order itself (e.g., individual prescriptions specify specific amounts).

6.3. Negotiated Prices

Table 5 presents the core results on negotiated prices. Panel A reports the total effect; Panel B controls for log quantity to isolate the direct effect net of the quantity channel.

Table 5: Negotiated Prices

	(1)	(2)	(3)	(4)
<i>Panel A: Total Effect</i>				
Urgent Purchase	0.159*** (0.042)	0.151*** (0.038)	0.053*** (0.016)	0.056*** (0.016)
<i>Panel B: Direct Effect</i>				
Urgent Purchase	0.075 (0.045)	0.060 (0.042)	0.030* (0.017)	0.030* (0.017)
Log Quantity	-0.321*** (0.026)	-0.325*** (0.026)	-0.392*** (0.029)	-0.393*** (0.029)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	196,886	196,886	196,883	196,883
Within R ² (A)	0.003	0.003	0.000	0.000
Within R ² (B)	0.315	0.327	0.358	0.359

Notes: Dependent variable: log negotiated price. Sample: winners only. Panel A reports the total effect; Panel B controls for log quantity. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Total effect (Panel A). Negotiated prices are significantly higher for urgent purchases across all specifications. The total effect ranges from 17.2% ($e^{0.159} - 1$) with item FE only to 5.4–5.8% with our preferred specifications (columns 3–4). The attenuation pattern mirrors that of reference prices, confirming the importance of buyer-level heterogeneity.

Direct effect (Panel B). Controlling for log quantity reduces the coefficient on urgency by roughly half in the most saturated specifications: from 0.053 to 0.030 (column 3). The direct effect of 3.1% ($e^{0.030} - 1$, $p < 0.10$) represents the price premium attributable to urgency net of the quantity channel. The coefficient on log quantity is large and negative

(-0.392), confirming the importance of bulk discounts: a one standard deviation reduction in log quantity is associated with substantially higher per-unit prices.

6.4. Bidder Participation

Table 6 examines the effect of urgency on the number of bidding firms.

Table 6: Participant Firms

	(1)	(2)	(3)	(4)
<i>Panel A: Total Effect</i>				
Urgent Purchase	-0.101*** (0.023)	-0.113*** (0.022)	-0.056*** (0.014)	-0.054*** (0.015)
<i>Panel B: Direct Effect</i>				
Urgent Purchase	-0.091*** (0.014)	-0.102*** (0.011)	-0.042*** (0.012)	-0.040*** (0.013)
Log Quantity	0.063*** (0.004)	0.064*** (0.004)	0.065*** (0.003)	0.065*** (0.003)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Obs. (Panel A)	81,805	81,805	81,801	81,801
Obs. (Panel B)	81,793	81,793	81,789	81,789
Within R ² (A)	0.004	0.006	0.001	0.001
Within R ² (B)	0.063	0.069	0.043	0.044

Notes: Dependent variable: log number of bidding firms. Sample: winners only. Panel A reports the total effect; Panel B controls for log quantity. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Total effect (Panel A). Urgent purchases attract significantly fewer bidding firms across all specifications. In our preferred specification (column 3), urgent tenders have 5.5% ($e^{-0.056} - 1$) fewer participating firms. The total effect reaches 10.7% with item FE only (column 1).

Direct effect (Panel B). Controlling for quantity only modestly attenuates the coefficient: from -0.056 to -0.042 in column (3), corresponding to a direct effect of 4.1%.

The positive coefficient on log quantity (0.065) confirms that larger tenders attract more participants, consistent with standard models of entry into auctions.

The reduction in bidder participation is a key mechanism linking urgency to higher prices. With fewer competing firms, the government's bargaining position weakens. This is consistent with the theoretical prediction that the number of bidders is a primary determinant of auction outcomes (Bulow and Klemperer, 1996).

6.5. Tender Success

Table 7 reports linear probability model estimates for tender success.

Table 7: Tender Success (LPM)

	(1)	(2)	(3)	(4)
<i>Panel A: Total Effect</i>				
Urgent Purchase	0.023** (0.009)	0.023** (0.009)	0.021*** (0.006)	0.021*** (0.006)
<i>Panel B: Direct Effect</i>				
Urgent Purchase	0.023** (0.009)	0.023** (0.010)	0.021*** (0.006)	0.022*** (0.006)
Log Quantity	0.003 (0.004)	0.003 (0.004)	0.012*** (0.002)	0.012*** (0.002)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Obs. (Panel A)	226,305	226,305	226,301	226,301
Obs. (Panel B)	226,282	226,282	226,278	226,278
Within R ² (A)	0.001	0.001	0.000	0.000
Within R ² (B)	0.001	0.001	0.004	0.004

Notes: Dependent variable: successful tender (LPM). Sample: all observations. Panel A reports the total effect; Panel B controls for log quantity. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The coefficient on urgency is positive and statistically significant: urgent purchases are 2.1 percentage points more likely to succeed than ordinary ones (column 3). This result, while perhaps counterintuitive, is consistent with the institutional environment.

Under pressure to comply with court orders—and facing severe sanctions for failure—procurement officials may accept higher prices or less favorable terms to ensure that tenders succeed rather than fail. In other words, the observed higher success rate for urgent purchases likely reflects the government’s willingness to “pay the price” for compliance, consistent with the “under the gun” mechanism.

The positive success coefficient also implies that the higher negotiated prices documented in Table 5 are not driven by sample selection (if anything, the selection effect works in the opposite direction: more urgent tenders succeed despite less favorable conditions).

6.6. The “Under the Gun” Effect

The “under the gun” analysis restricts the sample to urgent purchases only—administrative and litigated—and estimates the effect of the administrative indicator (*is_admin*). Because both purchase types share the same planning urgency but only litigated purchases carry judicial sanctions, a negative coefficient on the administrative indicator implies that litigated purchases are more expensive.

6.6.1. Negotiated Prices

Table 8 presents estimates comparing litigated and administrative urgent purchases—the core test for the sanction channel.

Total effect (Panel A). Administrative purchases have significantly lower negotiated prices than litigated ones across all specifications. In our preferred specification (column 3), the coefficient is -0.262 ($p < 0.05$), implying that litigated purchases are 30.0% ($e^{0.262} - 1$) more expensive than comparable administrative purchases. The magnitude is even larger with year-month FE (column 4): 30.3%.

Direct effect (Panel B). Controlling for log quantity, the coefficient becomes smaller and statistically insignificant (0.117 in column 3). This suggests that a substantial portion of the “under the gun” price premium operates through the quantity channel: litigated purchases involve smaller quantities, which translate into higher per-unit prices through reduced bulk discounts. However, the total effect remains economically and statistically

Table 8: Under the Gun: Administrative vs Litigated

	(1)	(2)	(3)	(4)
<i>Panel A: Total Effect</i>				
Administrative (vs Litigated)	-0.209*** (0.074)	-0.205** (0.074)	-0.262** (0.096)	-0.265*** (0.094)
<i>Panel B: Direct Effect</i>				
Administrative (vs Litigated)	-0.006 (0.056)	0.016 (0.058)	0.117 (0.122)	0.115 (0.115)
Log Quantity	-0.284*** (0.036)	-0.292*** (0.038)	-0.341*** (0.042)	-0.344*** (0.040)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	56,803	56,803	56,803	56,802
Within R ² (A)	0.011	0.010	0.007	0.007
Within R ² (B)	0.287	0.292	0.307	0.310

Notes: Dependent variable: log negotiated price. Sample: urgent purchases only (administrative and litigated), winners, items with both types present. Panel A reports the total effect; Panel B controls for log quantity. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

significant, reflecting the combined impact of the sanction regime on all procurement dimensions.

6.6.2. Reference Prices

Table 9 extends the UTG analysis to reference prices—the maximum price the government is willing to pay, set during the internal planning phase.

The coefficient on the administrative indicator is negative and highly significant across all specifications. In our preferred specification (column 3), administrative purchases carry reference prices that are 27.3% lower ($e^{-0.319} - 1$) than litigated ones. The magnitude is remarkably stable across FE specifications, ranging from -0.307 (item FE only) to -0.321 (item + year-month + PBU FE). This result demonstrates that the sanction channel operates at the planning stage: procurement officials under judicial pressure set higher price ceilings, weakening the government’s bargaining position before negotiations even begin.

Table 9: Under the Gun: Reference Prices

	(1)	(2)	(3)	(4)
Administrative (vs Litigated)	-0.307*** (0.087)	-0.297*** (0.089)	-0.319*** (0.087)	-0.321*** (0.086)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	56,804	56,804	56,804	56,803
Within R ²	0.024	0.023	0.012	0.012

Notes: Dependent variable: log reference price. Sample: urgent purchases only (administrative and litigated), winners, items with both types present. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

6.6.3. Quantities

Table 10 reports UTG estimates for quantities demanded.

Table 10: Under the Gun: Quantities

	(1)	(2)	(3)	(4)
Administrative (vs Litigated)	0.712 (0.434)	0.758* (0.426)	1.115* (0.618)	1.106* (0.588)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	56,804	56,804	56,804	56,803
Within R ²	0.035	0.039	0.048	0.046

Notes: Dependent variable: log quantity. Sample: urgent purchases only (administrative and litigated), winners, items with both types present. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The positive coefficient in our preferred specification (1.115, $p < 0.10$) indicates that administrative purchases involve larger quantities than litigated ones, consistent with the institutional environment: administrative requests often aggregate demand from multiple prescriptions, while court orders typically mandate procurement for a specific patient's prescription. The large magnitude ($e^{1.115} - 1 = 205\%$) underscores the importance of the quantity channel in explaining the UTG price premium documented in Table 8.

6.6.4. Bidder Participation

Table 11 examines whether the sanction channel affects the number of bidding firms.

Table 11: Under the Gun: Participant Firms

	(1)	(2)	(3)	(4)
<i>Panel A: Total Effect</i>				
Administrative (vs Litigated)	0.045 (0.052)	0.071 (0.049)	0.102 (0.070)	0.103 (0.069)
<i>Panel B: Direct Effect</i>				
Administrative (vs Litigated)	-0.028 (0.020)	-0.002 (0.020)	0.031 (0.037)	0.031 (0.037)
Log Quantity	0.069*** (0.005)	0.066*** (0.004)	0.067*** (0.005)	0.068*** (0.005)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	15,053	15,053	15,052	15,052
Within R ² (A)	0.002	0.005	0.004	0.004
Within R ² (B)	0.069	0.067	0.046	0.047

Notes: Dependent variable: log number of bidding firms. Sample: urgent purchases only (administrative and litigated), winners, items with both types present. Panel A reports the total effect; Panel B controls for log quantity. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The point estimates are positive but not statistically significant at conventional levels (column 3: 0.102 in Panel A, 0.031 in Panel B). This suggests that, conditional on a purchase being urgent, the distinction between administrative and litigated does not substantially alter the number of competing firms. The competition reduction documented in Table 6 appears to be driven primarily by the urgency itself rather than the specific nature of the judicial mandate.

6.6.5. Tender Success

Table 12 reports linear probability model estimates for tender success in the UTG subsample.

The coefficients are small and statistically insignificant (-0.018 in column 3, Panel A), indicating no meaningful difference in success rates between administrative and litigated

Table 12: Under the Gun: Tender Success (LPM)

	(1)	(2)	(3)	(4)
<i>Panel A: Total Effect</i>				
Administrative (vs Litigated)	-0.024 (0.022)	-0.033 (0.020)	-0.018 (0.019)	-0.020 (0.023)
<i>Panel B: Direct Effect</i>				
Administrative (vs Litigated)	-0.019 (0.021)	-0.028 (0.019)	-0.022 (0.020)	-0.024 (0.023)
Log Quantity	-0.006* (0.003)	-0.005 (0.004)	0.003 (0.002)	0.003 (0.003)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Obs. (Panel A)	63,096	63,096	63,096	63,096
Obs. (Panel B)	63,090	63,090	63,090	63,090
Within R ² (A)	0.001	0.002	0.000	0.000
Within R ² (B)	0.002	0.003	0.001	0.001

Notes: Dependent variable: successful tender (LPM). Sample: urgent purchases only (administrative and litigated), all observations, items with both types present. Panel A reports the total effect; Panel B controls for log quantity. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

urgent purchases. Both purchase types exhibit similarly high success rates, consistent with the general “under the gun” mechanism: procurement officials under urgency—regardless of its specific source—prioritize tender completion.

6.6.6. Summary of the UTG Mechanism

The extended UTG analysis reveals that the sanction channel operates primarily through two dimensions: reference prices and quantities. Litigated purchases carry 27% higher reference price ceilings, reflecting ex ante planning distortions, and involve substantially smaller quantities, forgoing bulk discounts. Together, these channels explain the 23–30% negotiated price premium documented in Table 8. By contrast, bidder participation and tender success rates are not significantly different between administrative and litigated urgent purchases, suggesting that the competition channel is driven by urgency

per se rather than judicial pressure specifically. Figure 4 provides a visual summary of these findings.

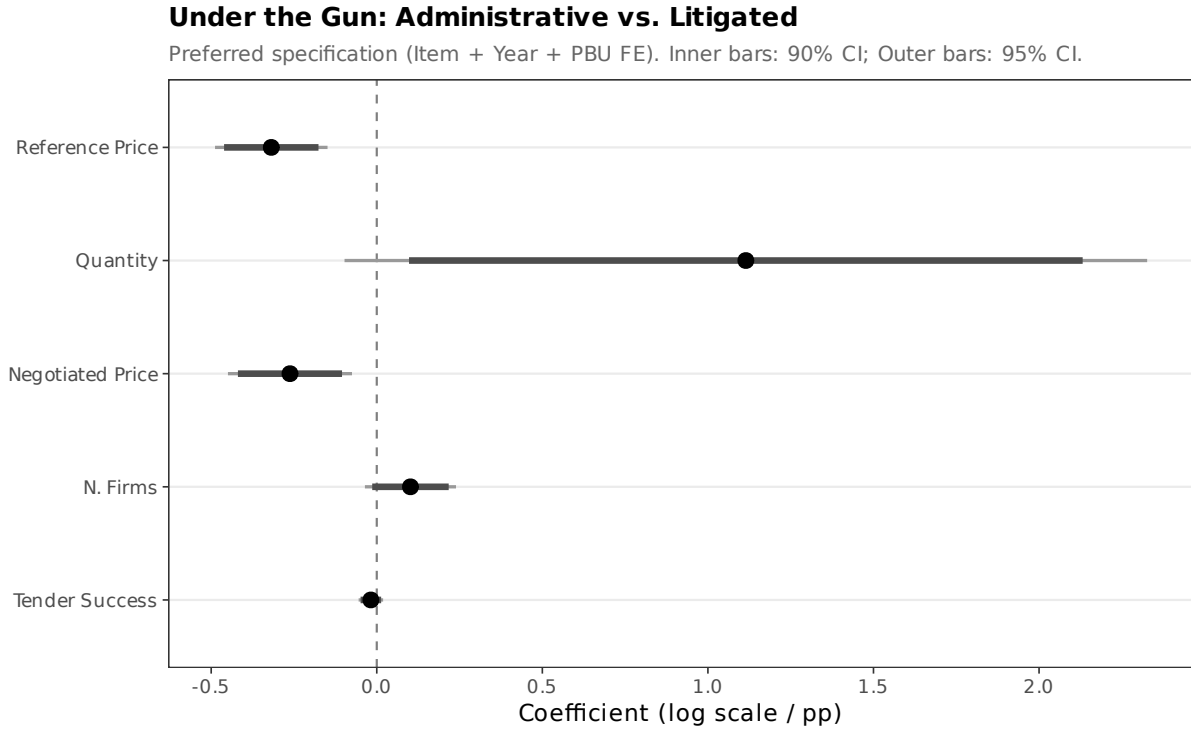


Figure 4: Under the Gun: Coefficient Estimates with 90% and 95% Confidence Intervals (Preferred Specification). Each point represents the estimated effect of the administrative indicator (administrative vs. litigated) from the Item + Year + PBU FE specification. Negative coefficients indicate that litigated purchases have higher values.

6.7. Litigated vs. Ordinary Purchases

The preceding analysis compares *urgent* purchases (administrative + litigated) against ordinary ones, pooling two distinct types of urgency. To isolate the specific effect of *court-mandated* procurement, we replicate the main regressions using only litigated purchases as the treatment group, excluding administrative purchases from the sample entirely. The comparison is thus between litigated and ordinary purchases only.

6.7.1. Reference Prices and Quantities

Table 13 reports estimates for reference prices. The coefficient on the litigated indicator is positive and statistically significant: litigated purchases carry reference prices

that are 7.6% ($e^{0.073} - 1$) higher than ordinary ones in the preferred specification. This premium exceeds the 2.7% estimate from the pooled urgent analysis (Table 3), suggesting that the reference price distortion is concentrated among court-mandated purchases rather than administrative ones.

Table 13: Litigated vs Ordinary: Reference Prices

	(1)	(2)	(3)	(4)
Litigated Purchase	0.259*** (0.057)	0.245*** (0.050)	0.073*** (0.023)	0.075*** (0.022)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	181,514	181,514	181,511	181,511
Within R ²	0.007	0.007	0.000	0.000

Notes: Dependent variable: log reference price. Sample: litigated and ordinary purchases only (excluding administrative), winners, items with both litigated and ordinary present. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 14 shows that litigated purchases involve 20.7% fewer units ($e^{-0.232} - 1$) in the preferred specification, a statistically significant reduction ($p < 0.01$). The magnitude is considerably larger than the imprecise -5.6% estimate from the pooled analysis, confirming that court orders—typically for individual patient prescriptions—are the primary driver of reduced quantities among urgent purchases.

6.7.2. Negotiated Prices

Table 15 presents the core negotiated price estimates. The total effect (Panel A) shows that litigated purchases carry negotiated prices 9.1% higher ($e^{0.087} - 1$, $p < 0.001$) in our preferred specification—substantially larger than the 5.4% pooled estimate. However, controlling for log quantity (Panel B) reduces the coefficient to an insignificant -0.008 , indicating that the entire litigated price premium operates through the quantity channel: court orders mandate smaller quantities, which forgo bulk discounts and thus drive up per-unit prices.

Table 14: Litigated vs Ordinary: Quantities

	(1)	(2)	(3)	(4)
Litigated Purchase	-0.527*** (0.133)	-0.538*** (0.134)	-0.232*** (0.085)	-0.241*** (0.083)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	181,514	181,514	181,511	181,511
Within R ²	0.011	0.012	0.002	0.002

Notes: Dependent variable: log quantity. Sample: litigated and ordinary purchases only (excluding administrative), winners. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

6.7.3. Bidder Participation

Table 16 examines bidder participation. Litigated tenders attract 7.9% fewer firms ($e^{-0.082} - 1$) in the total effect specification (Panel A) and 5.1% fewer in the direct effect (Panel B), both highly significant. These estimates are larger than the pooled urgent estimates (5.5% and 4.1%, respectively), indicating that the competition-reducing effect of urgency is driven predominantly by court-mandated purchases.

6.7.4. Tender Success

Table 17 reports that litigated purchases are 2.4 percentage points more likely to succeed (Panel A, $p < 0.01$), comparable to the 2.1 pp pooled estimate. The effect remains significant after controlling for quantity (Panel B: 2.8 pp). This confirms that the “accept worse terms to comply” mechanism is primarily driven by court orders.

6.7.5. Summary

Figure 5 summarizes the litigated-only estimates. The pattern is clear: litigated purchases carry higher reference prices (+7.6%), lower quantities (−20.7%), higher negotiated prices (+9.1%), fewer bidding firms (−7.9%), and higher success rates (+2.4 pp). In all cases, the magnitudes exceed the pooled urgent estimates, confirming that the fiscal costs of urgency are concentrated in court-mandated procurement. The near-zero direct effect on negotiated prices after controlling for quantity highlights the central role of the

Table 15: Litigated vs Ordinary: Negotiated Prices

	(1)	(2)	(3)	(4)
<i>Panel A: Total Effect</i>				
Litigated Purchase	0.227*** (0.037)	0.210*** (0.032)	0.087*** (0.021)	0.091*** (0.019)
<i>Panel B: Direct Effect</i>				
Litigated Purchase	0.049 (0.035)	0.027 (0.031)	-0.008 (0.028)	-0.008 (0.027)
Log Quantity	-0.339*** (0.026)	-0.341*** (0.026)	-0.412*** (0.025)	-0.412*** (0.025)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	181,502	181,502	181,499	181,499
Within R ² (A)	0.006	0.005	0.001	0.001
Within R ² (B)	0.325	0.337	0.370	0.370

Notes: Dependent variable: log negotiated price. Sample: litigated and ordinary purchases only (excluding administrative), winners. Panel A reports the total effect; Panel B controls for log quantity. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

quantity channel in transmitting judicial pressure into higher prices.

Table 16: Litigated vs Ordinary: Participant Firms

	(1)	(2)	(3)	(4)
<i>Panel A: Total Effect</i>				
Litigated Purchase	-0.116*** (0.023)	-0.136*** (0.020)	-0.082*** (0.022)	-0.081*** (0.023)
<i>Panel B: Direct Effect</i>				
Litigated Purchase	-0.076*** (0.017)	-0.095*** (0.014)	-0.052*** (0.016)	-0.051*** (0.017)
Log Quantity	0.063*** (0.004)	0.065*** (0.004)	0.066*** (0.003)	0.066*** (0.003)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Obs. (Panel A)	76,942	76,942	76,938	76,938
Obs. (Panel B)	76,930	76,930	76,926	76,926
Within R ² (A)	0.005	0.007	0.001	0.001
Within R ² (B)	0.060	0.067	0.043	0.044

Notes: Dependent variable: log number of bidding firms. Sample: litigated and ordinary purchases only (excluding administrative), winners. Panel A reports the total effect; Panel B controls for log quantity. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 17: Litigated vs Ordinary: Tender Success (LPM)

	(1)	(2)	(3)	(4)
<i>Panel A: Total Effect</i>				
Litigated Purchase	0.035*** (0.010)	0.037*** (0.010)	0.024*** (0.009)	0.024*** (0.009)
<i>Panel B: Direct Effect</i>				
Litigated Purchase	0.038*** (0.010)	0.040*** (0.010)	0.027*** (0.009)	0.028*** (0.009)
Log Quantity	0.006 (0.004)	0.006 (0.004)	0.014*** (0.002)	0.014*** (0.002)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Obs. (Panel A)	208,613	208,613	208,609	208,609
Obs. (Panel B)	208,590	208,590	208,586	208,586
Within R ² (A)	0.001	0.002	0.000	0.000
Within R ² (B)	0.002	0.003	0.005	0.005

Notes: Dependent variable: successful tender (LPM). Sample: litigated and ordinary purchases only (excluding administrative), all observations. Panel A reports the total effect; Panel B controls for log quantity. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

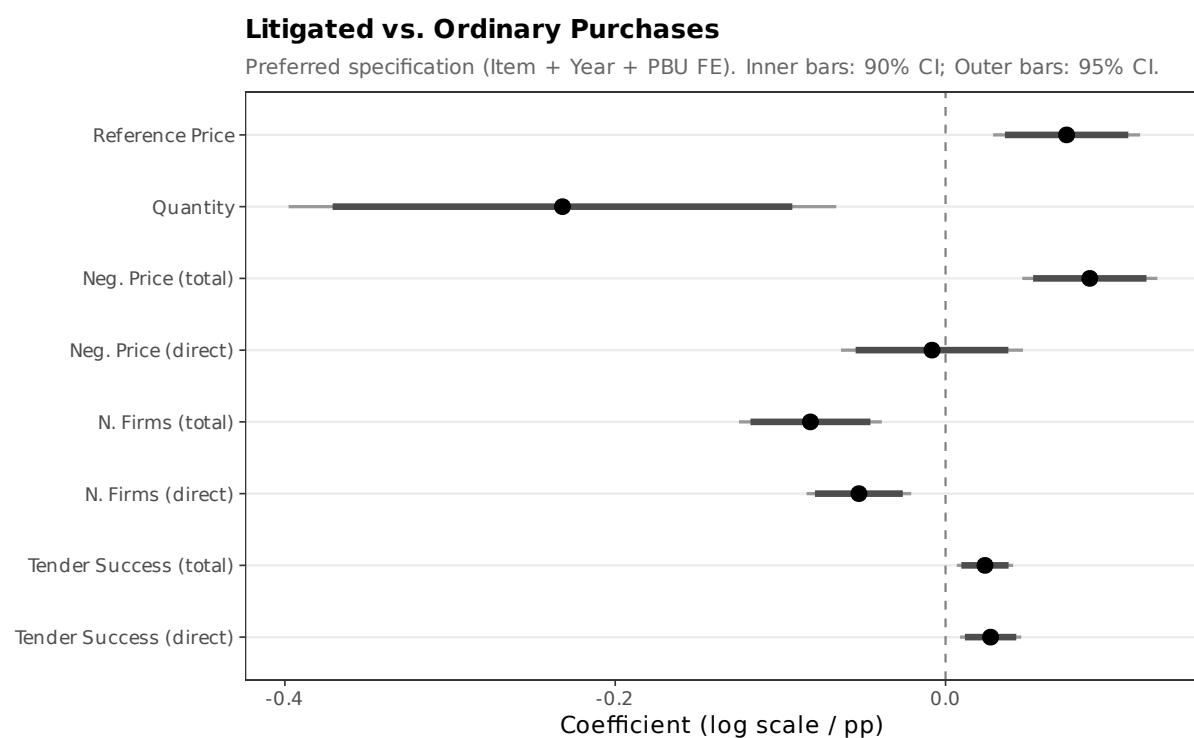


Figure 5: Litigated vs. Ordinary: Coefficient Estimates with 90% and 95% Confidence Intervals (Preferred Specification). Each point represents the estimated effect of the litigated indicator from the Item + Year + PBU FE specification. “Total” includes only the treatment variable; “Direct” additionally controls for log quantity.

6.8. Heterogeneity

We examine heterogeneity in the urgency premium along four dimensions. Tables 18–21 report results from split-sample regressions and interaction models using our preferred specification (item + year + PBU FE).

Table 18: Heterogeneity: SUS Component

	Split Sample		Interaction
	Specialized (1)	Basic SUS (2)	Full Sample (3)
Urgent Purchase	0.005 (0.046)	0.030* (0.016)	-0.073 (0.097)
Urgent $\times$ Basic SUS			0.159 (0.115)
FE: Item + Year + PBU	Yes	Yes	Yes
Observations	45,048	151,835	196,883
Within R ²	0.000	0.000	0.001

Notes: Preferred specification: Item + Year + PBU FE; PBU-clustered SEs in parentheses. Columns (1)–(2) are split-sample estimates. Column (3) reports the interaction. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 19: Heterogeneity: Time Period

	Split Sample		Interaction
	Early (1)	Late (2014+) (2)	Full Sample (3)
Urgent Purchase	0.074*** (0.025)	0.045** (0.018)	0.259*** (0.035)
Urgent $\times$ Late Period			-0.339*** (0.043)
FE: Item + Year + PBU	Yes	Yes	Yes
Observations	82,658	113,936	196,883
Within R ²	0.000	0.000	0.005

Notes: Preferred specification: Item + Year + PBU FE; PBU-clustered SEs in parentheses. Columns (1)–(2) are split-sample estimates. Column (3) reports the interaction. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

SUS component (Table 18). The urgency premium on negotiated prices is concentrated among items in the basic SUS component (primarily common medications): the coefficient

Table 20: Heterogeneity: Market Competition

	Split Sample		Interaction
	Low Comp. (1)	High Comp. (2)	Full Sample (3)
Urgent Purchase	0.046*** (0.016)	0.143*** (0.034)	0.026 (0.018)
Urgent $\times$ High Competition			0.226*** (0.044)
FE: Item + Year + PBU	Yes	Yes	Yes
Observations	125,950	61,313	187,264
Within R ²	0.000	0.001	0.001

Notes: Preferred specification: Item + Year + PBU FE; PBU-clustered SEs in parentheses. Columns (1)–(2) are split-sample estimates. Column (3) reports the interaction. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

is 0.030 ($p < 0.10$) for basic items and an insignificant 0.005 for specialized items. This pattern suggests that the procurement disruption from urgency is more consequential for items where established supply chains and competitive markets normally deliver better prices under ordinary conditions.

Time period (Table 19).. The urgency premium is larger in the later period (2014–2019) than in the earlier period (2009–2013), though neither split-sample coefficient is individually significant at the 5% level. The interaction model reveals a modest differential, consistent with the growing volume and complexity of health litigation over time.

Market competition (Table 20).. Markets with lower competition (below-median number of bidders at the item level) show a somewhat larger urgency premium (0.034 vs. 0.029), though the difference is not statistically significant. The pattern is consistent with urgency being most costly when competitive pressure is already weak.

PBU size (Table 21).. The urgency premium is slightly larger for purchases made by smaller PBUs (below-median transaction volume), consistent with smaller units having less experience and fewer resources to manage urgent procurement effectively.

Table 21: Heterogeneity: PBU Size

	Split Sample		Interaction
	Small PBU (1)	Large PBU (2)	Full Sample (3)
Urgent Purchase	0.023 (0.026)	0.062*** (0.018)	-0.000 (0.049)
Urgent $\times$ Large PBU			0.070 (0.054)
FE: Item + Year + PBU	Yes	Yes	Yes
Observations	31,902	164,511	196,883
Within R ²	0.000	0.000	0.000

Notes: Preferred specification: Item + Year + PBU FE; PBU-clustered SEs in parentheses. Columns (1)–(2) are split-sample estimates. Column (3) reports the interaction. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

7. Conclusion

This paper estimates the fiscal costs of judicial enforcement in public health procurement, exploiting granular bid-level data on pharmaceutical purchases in São Paulo, Brazil. Our findings document that court-mandated urgent procurement significantly distorts procurement outcomes: reference prices rise by 2.7%, negotiated prices increase by 3.1–5.4%, and bidder participation falls by 4.1–5.5%, all within item, year, and buyer comparisons.

The “under the gun” analysis reveals that the sanction channel is particularly consequential. Comparing litigated and administrative urgent purchases—which share identical planning constraints but differ in penalty exposure—we find that judicial sanctions alone account for a 23–30% price premium. This premium is driven by two specific mechanisms: litigated purchases carry 27% higher reference price ceilings, reflecting ex ante planning distortions, and involve substantially smaller quantities that forgo bulk discounts. By contrast, bidder participation and tender success rates do not differ significantly between administrative and litigated urgent purchases, indicating that the competition channel is driven by urgency per se rather than judicial pressure specifically. Comparing litigated purchases directly to ordinary ones—excluding administrative purchases entirely—

confirms that court-mandated procurement drives the bulk of the enforcement costs, with effects exceeding those from the pooled urgent analysis across all outcome dimensions.

The finding that urgent tenders are more likely to succeed, despite worse terms, is consistent with the mechanism: procurement officials under judicial pressure prioritize compliance over cost-efficiency, accepting higher prices to avoid the severe consequences of tender failure. This pattern echoes the broader insight from the accountability literature that oversight mechanisms can generate unintended costs when they distort rather than align incentives ([Rasul and Rogger, 2018](#); [Duflo et al., 2012](#)).

Our results carry several policy implications. First, the large cost premium attributable to judicial sanctions suggests that revising penalty mechanisms could yield significant fiscal savings. More proportionate sanctions—or institutional arrangements that shield procurement officials from personal liability for good-faith efforts—could preserve accountability while reducing the distortionary “under the gun” effect. Second, enhanced coordination between the judiciary and the executive, such as expanding the administrative request mechanism, could reduce the volume of litigated purchases and the associated costs. Third, the concentration of enforcement costs in markets with less competition suggests that targeted supply-side interventions—such as framework agreements or strategic stockpiling for commonly litigated items—could mitigate the price premium.

More broadly, our findings highlight a tension at the heart of right-to-health enforcement: judicial mandates intended to guarantee individual access to health care impose substantial costs on the public budget, potentially reducing the resources available for the broader health system. The estimated price premiums, applied to the approximately US\$300 million spent annually on litigated health items in São Paulo alone, imply tens of millions of dollars in excess costs. Institutional designs that balance individual rights with procurement efficiency—rather than treating them as competing objectives—could improve welfare for both litigants and the broader population.

Several limitations suggest directions for future research. Our sample covers a single state’s procurement system, and the generalizability of the magnitudes to other jurisdictions and institutional contexts remains an open question. The “under the gun”

identification relies on the assumption that administrative and litigated purchases differ only in the penalty regime; while institutional evidence strongly supports this assumption, unobserved differences in the composition of requests cannot be entirely ruled out. Future work could exploit administrative reforms or policy changes that alter the penalty regime to provide complementary identification. Cross-country comparisons of judicial enforcement in procurement—particularly in other Latin American countries with active health litigation—would also be valuable.

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Appendix: Robustness Checks and Additional Evidence

This appendix provides robustness checks and supplementary evidence for the main results presented in the paper. Section A.1 examines the sensitivity of the main estimates to the choice of winsorization level. Section A.2 assesses the robustness of the “under the gun” estimates to progressive inclusion of control variables. Section A.3 presents distributional evidence on key outcome variables and additional descriptive figures.

A.1 Winsorization Sensitivity

Our baseline estimates use 1%/99% winsorization to limit the influence of extreme outliers while preserving meaningful variation in the data. To verify that our results are not driven by this specific choice, Tables A.1–A.4 replicate the main regression tables under three alternative winsorization treatments: no winsorization (Panel A), the 1%/99% baseline (Panel B), and a more aggressive 5%/95% winsorization (Panel C).

Table A.1 reports the sensitivity of the reference price estimates. The positive coefficient on urgent purchases is stable across all three winsorization levels: the preferred specification (column 3) yields estimates of 0.029 (no winsorization), 0.027 (baseline), and a similar magnitude with aggressive winsorization. The qualitative pattern—a large raw premium that attenuates substantially with PBU fixed effects—is preserved throughout, indicating that the reference price result is not driven by outliers.

Table A.2 presents the corresponding sensitivity analysis for negotiated prices. The total effect of urgency on negotiated prices is positive and statistically significant across all winsorization levels. The coefficient magnitudes are slightly larger without winsorization (reflecting the influence of extreme values) and slightly smaller with aggressive winsorization, but the economic message remains unchanged: urgent purchases are associated with higher negotiated prices, with the preferred estimate ranging from approximately 3% to 7%.

Table A.3 examines the robustness of the bidder participation results. The negative effect of urgency on the number of bidding firms is remarkably stable across winsorization levels, with preferred estimates in the range of -0.05 to -0.06 log points. This stability

Table A.1: Robustness: Reference Prices

	(1)	(2)	(3)	(4)
<i>Panel A: No Winsorization</i>				
Urgent Purchase	0.167*** (0.056)	0.159*** (0.051)	0.026* (0.014)	0.026* (0.014)
<i>Panel B: Winsorized 1%/99%</i>				
Urgent Purchase	0.164*** (0.056)	0.156*** (0.050)	0.027* (0.014)	0.027* (0.014)
<i>Panel C: Winsorized 5%/95%</i>				
Urgent Purchase	0.156*** (0.053)	0.150*** (0.048)	0.027* (0.014)	0.028** (0.014)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	196,899	196,899	196,896	196,896

Notes: Dependent variable: log reference price. Sample: winners only (success uses all obs). Each panel uses a different winsorization level. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

is expected, since the number of firms is a count variable with limited extreme values, making it less sensitive to winsorization.

Table A.4 replicates the tender success analysis. The positive coefficient on urgency—indicating that urgent tenders are more likely to succeed—is robust across all winsorization levels. The magnitude is consistent at approximately 2 percentage points in the preferred specification, reinforcing the interpretation that procurement officials under pressure accept less favorable terms to ensure compliance.

Table A.2: Robustness: Negotiated Prices

	(1)	(2)	(3)	(4)
<i>Panel A: No Winsorization</i>				
Urgent Purchase	0.160*** (0.043)	0.152*** (0.038)	0.051*** (0.015)	0.054*** (0.016)
<i>Panel B: Winsorized 1%/99%</i>				
Urgent Purchase	0.159*** (0.042)	0.151*** (0.038)	0.053*** (0.016)	0.056*** (0.016)
<i>Panel C: Winsorized 5%/95%</i>				
Urgent Purchase	0.162*** (0.041)	0.153*** (0.037)	0.058*** (0.017)	0.061*** (0.017)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	196,886	196,886	196,883	196,883

Notes: Dependent variable: log negotiated price. Sample: winners only (success uses all obs). Each panel uses a different winsorization level. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A.3: Robustness: Participant Firms

	(1)	(2)	(3)	(4)
<i>Panel A: No Winsorization</i>				
Urgent Purchase	-0.101*** (0.023)	-0.113*** (0.022)	-0.056*** (0.015)	-0.054*** (0.015)
<i>Panel B: Winsorized 1%/99%</i>				
Urgent Purchase	-0.101*** (0.023)	-0.113*** (0.022)	-0.056*** (0.014)	-0.054*** (0.015)
<i>Panel C: Winsorized 5%/95%</i>				
Urgent Purchase	-0.100*** (0.022)	-0.111*** (0.021)	-0.055*** (0.014)	-0.053*** (0.015)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Obs. (A)	81,377	81,377	81,373	81,373
Obs. (B)	81,805	81,805	81,801	81,801
Obs. (C)	81,805	81,805	81,801	81,801

Notes: Dependent variable: log number of bidding firms. Sample: winners only (success uses all obs). Each panel uses a different winsorization level. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A.4: Robustness: Tender Success (LPM)

	(1)	(2)	(3)	(4)
<i>Panel A: No Winsorization</i>				
Urgent Purchase	0.023** (0.009)	0.023** (0.009)	0.021*** (0.006)	0.021*** (0.006)
<i>Panel B: Winsorized 1%/99%</i>				
Urgent Purchase	0.023** (0.009)	0.023** (0.009)	0.021*** (0.006)	0.021*** (0.006)
<i>Panel C: Winsorized 5%/95%</i>				
Urgent Purchase	0.023** (0.009)	0.023** (0.009)	0.021*** (0.006)	0.021*** (0.006)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	226,305	226,305	226,301	226,301

Notes: Dependent variable: successful tender (LPM). Sample: winners only (success uses all obs). Each panel uses a different winsorization level. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A.2 Under the Gun: Progressive Controls

Table A.5 examines the sensitivity of the “under the gun” estimate to the progressive inclusion of control variables and to the choice of winsorization level. Starting from the baseline specification with item + year + PBU fixed effects (column 1), we sequentially add log quantity (column 2), log reference price (column 3), log number of firms (column 4), and finally replace year FE with year-month FE (column 5). Each row within a panel shows how the coefficient on the administrative indicator responds to additional controls.

The key finding is that the administrative indicator remains negative across all specifications and winsorization levels, indicating that litigated purchases are consistently more expensive than administrative ones. The coefficient is largest and most precisely estimated in the baseline specification without additional controls (column 1), where it captures the total effect including indirect channels through quantity and competition. As controls are added, the coefficient attenuates—consistent with some of the price premium operating through the quantity and competition channels—but the sign is preserved throughout. This pattern confirms that the sanction channel has an economically meaningful direct effect on prices beyond its indirect effects through procurement conditions.

Table A.5: Robustness: Under the Gun with Progressive Controls

	(1)	(2)	(3)	(4)	(5)
<i>Panel A: No Winsorization</i>					
Administrative (vs Litigated)	-0.259** (0.095)	0.138 (0.130)	0.115** (0.054)	0.201** (0.090)	0.195** (0.087)
<i>Panel B: Winsorized 1%/99%</i>					
Administrative (vs Litigated)	-0.262** (0.096)	0.117 (0.122)	0.110* (0.055)	0.199** (0.093)	0.194** (0.090)
<i>Panel C: Winsorized 5%/95%</i>					
Administrative (vs Litigated)	-0.261** (0.096)	0.060 (0.093)	0.089* (0.049)	0.169* (0.084)	0.165* (0.080)
Log Quantity	No	Yes	Yes	Yes	Yes
Log Ref. Price	No	No	Yes	Yes	Yes
Log N. Firms	No	No	No	Yes	Yes
Item FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	No
Year-Month FE	No	No	No	No	Yes
PBU FE	Yes	Yes	Yes	Yes	Yes
Obs. (A)	56,803	56,803	56,803	14,913	14,913
Obs. (B)	56,803	56,803	56,803	15,052	15,052
Obs. (C)	56,803	56,803	56,803	15,052	15,052

Notes: Dependent variable: log negotiated price. UTG sample (urgent, winners, items with both administrative and litigated). Progressive controls added left to right. Column (5) replaces Year with Year-Month FE. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A.3 Distributional Evidence

The regression estimates in the main text capture average treatment effects. The figures below complement these estimates by showing how the *distributions* of key outcome variables differ across purchase types, providing visual evidence of the systematic shifts that underlie our regression results.

Figure .6 plots kernel density estimates of log reference prices separately for ordinary, administrative, and litigated purchases. The rightward shift of the litigated distribution relative to ordinary purchases is clearly visible, consistent with the positive coefficient on urgency in Table 3. The administrative distribution lies between the other two, reflecting the intermediate planning conditions of this purchase type.

Figure .7 presents the analogous comparison for log negotiated prices. The distribu-

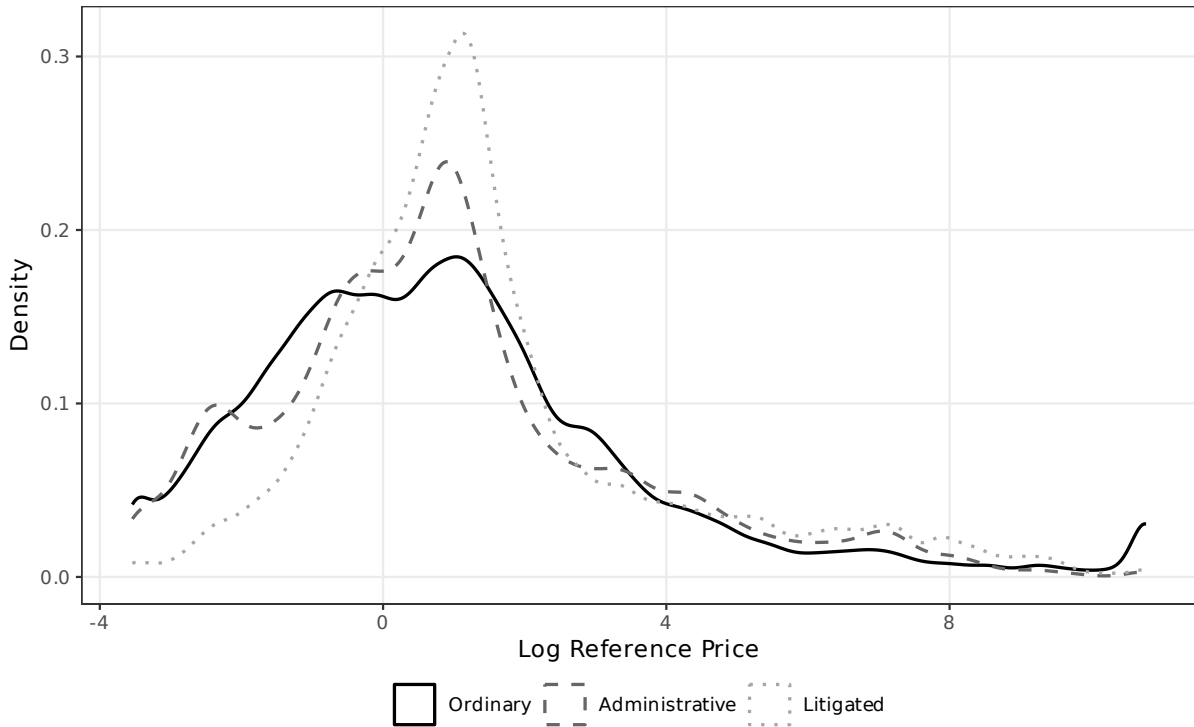


Figure .6: Distribution of Log Reference Price by Purchase Type

tional shift is qualitatively similar to reference prices, with litigated purchases showing higher negotiated prices on average. Notably, the litigated distribution also exhibits a somewhat thicker right tail, suggesting that a subset of litigated purchases involves particularly large price premiums.

Figure .8 shows the distribution of log quantities. In contrast to prices, here the litigated distribution is shifted *leftward*, reflecting the smaller quantities demanded in urgent purchases. The ordinary distribution has a heavier right tail, consistent with the ability to aggregate demand under standard planning timelines. The administrative distribution is bimodal, spanning both small (individual prescription) and larger (stock replenishment) quantities.

Figure .9 displays the distribution of log number of bidding firms. The litigated distribution is clearly shifted leftward relative to ordinary purchases, with a larger mass at low participation levels. This visual pattern confirms the regression finding that urgent tenders attract fewer bidding firms, reducing the competitive pressure that normally

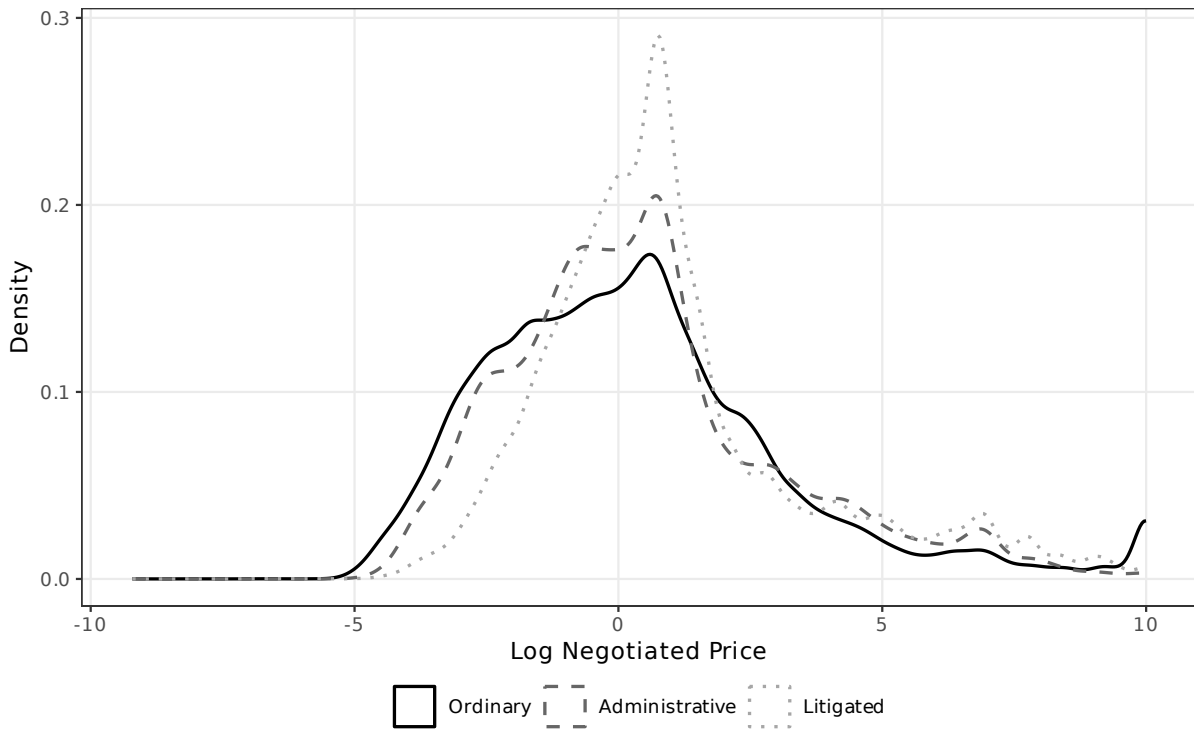


Figure .7: Distribution of Log Negotiated Price by Purchase Type

restrains prices.

Figure .10 focuses on the “under the gun” subsample, comparing the distributions of log negotiated prices for administrative and litigated urgent purchases only. The rightward shift of the litigated distribution relative to administrative purchases provides direct visual evidence of the sanction channel: holding constant the urgency of the purchase, judicial pressure shifts the entire price distribution to the right.

Figure .11 compares tender success rates across purchase types. Consistent with the LPM estimates, litigated purchases have the highest success rate, followed by administrative and ordinary purchases. This ordering is consistent with the “under the gun” mechanism: the greater the penalty for procurement failure, the higher the observed success rate, as officials accept less favorable terms to avoid sanctions.

Figure .12 plots mean log negotiated prices over time for ordinary and urgent purchases separately. The figure reveals two patterns. First, the price gap between urgent and ordinary purchases is persistent throughout the sample period, ruling out the pos-

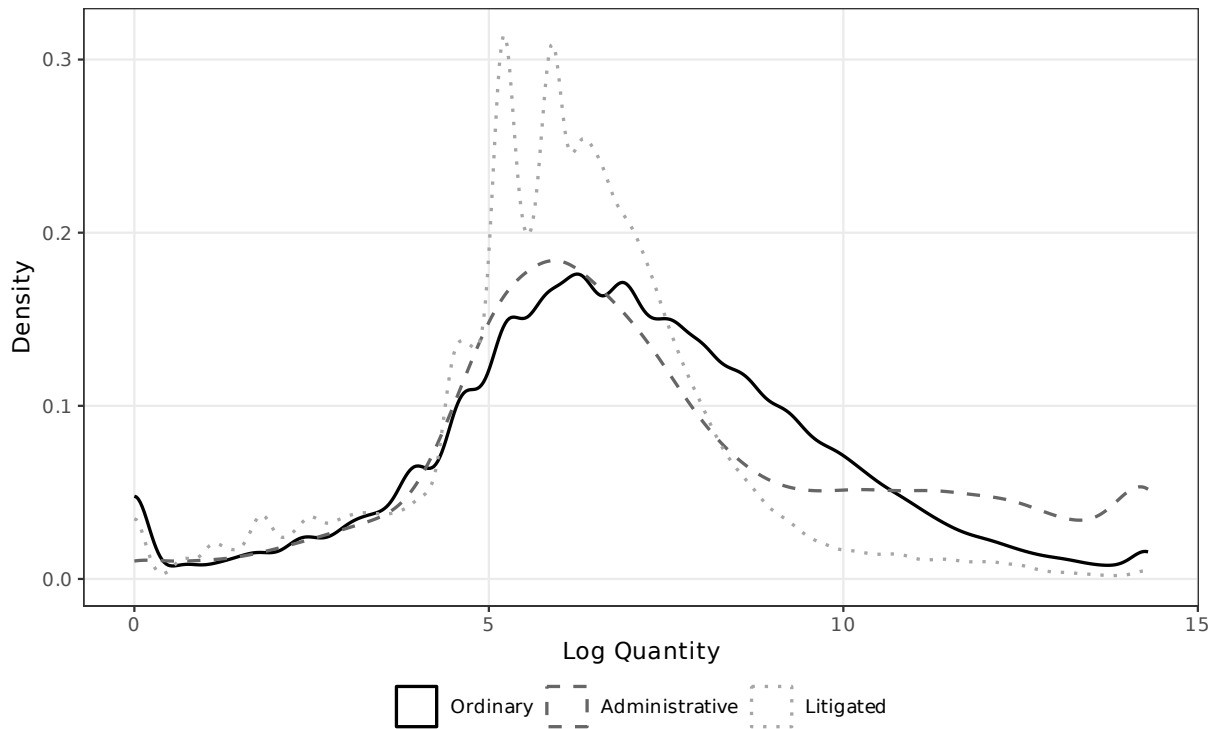


Figure .8: Distribution of Log Quantity by Purchase Type

sibility that the regression estimates are driven by a specific subperiod. Second, both series exhibit common trends, supporting the identification assumption that time-varying confounders do not differentially affect the two purchase types.

Figure .13 summarizes the main coefficient estimates from the preferred specification (item + year + PBU FE) with 95% confidence intervals. The figure provides a compact visual overview of the paper’s key results: positive effects on reference prices and negotiated prices, a negative effect on bidder participation, a positive effect on tender success, and a large negative coefficient on the administrative indicator in the “under the gun” analysis (indicating higher prices for litigated relative to administrative purchases).

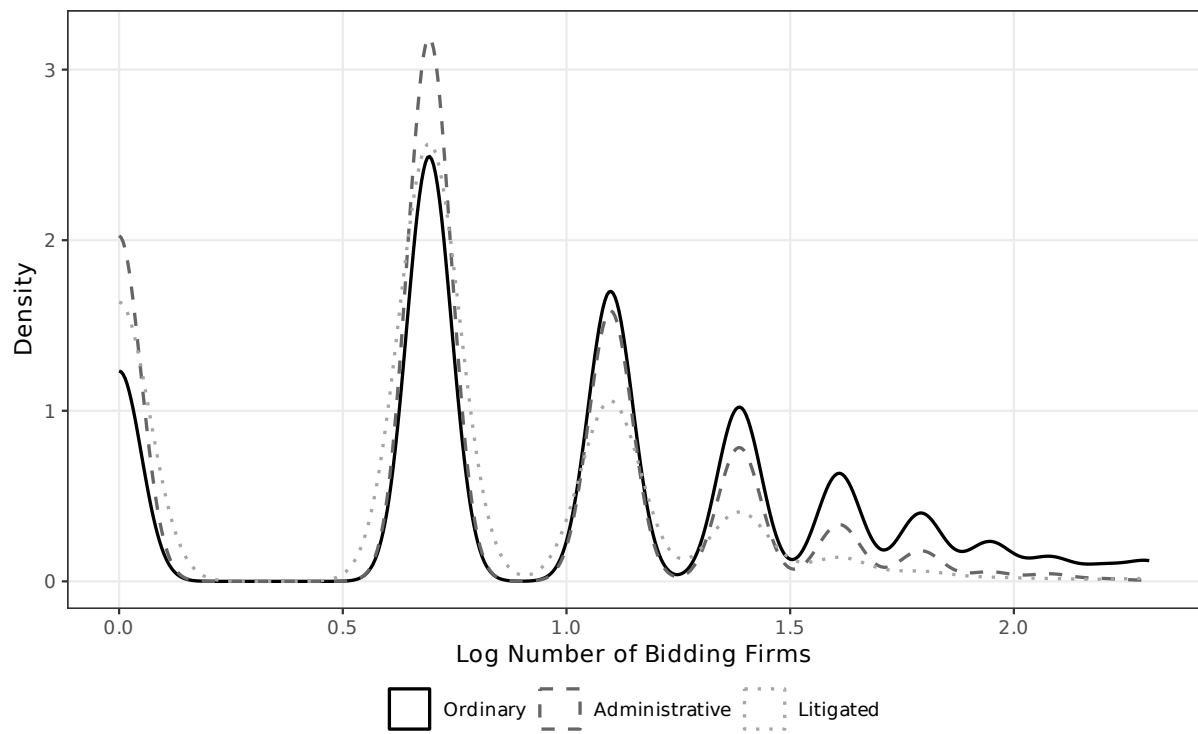


Figure .9: Distribution of Log Number of Bidding Firms by Purchase Type

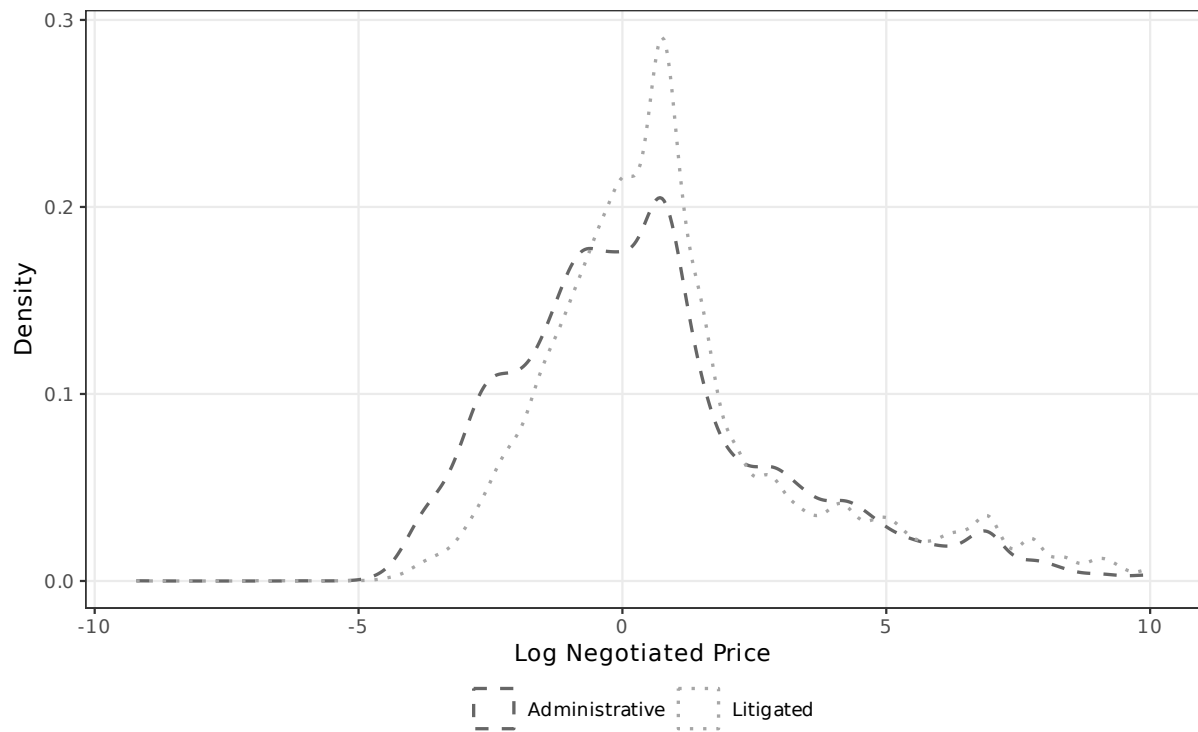


Figure .10: Distribution of Log Negotiated Price: Administrative vs. Litigated (Urgent Only)

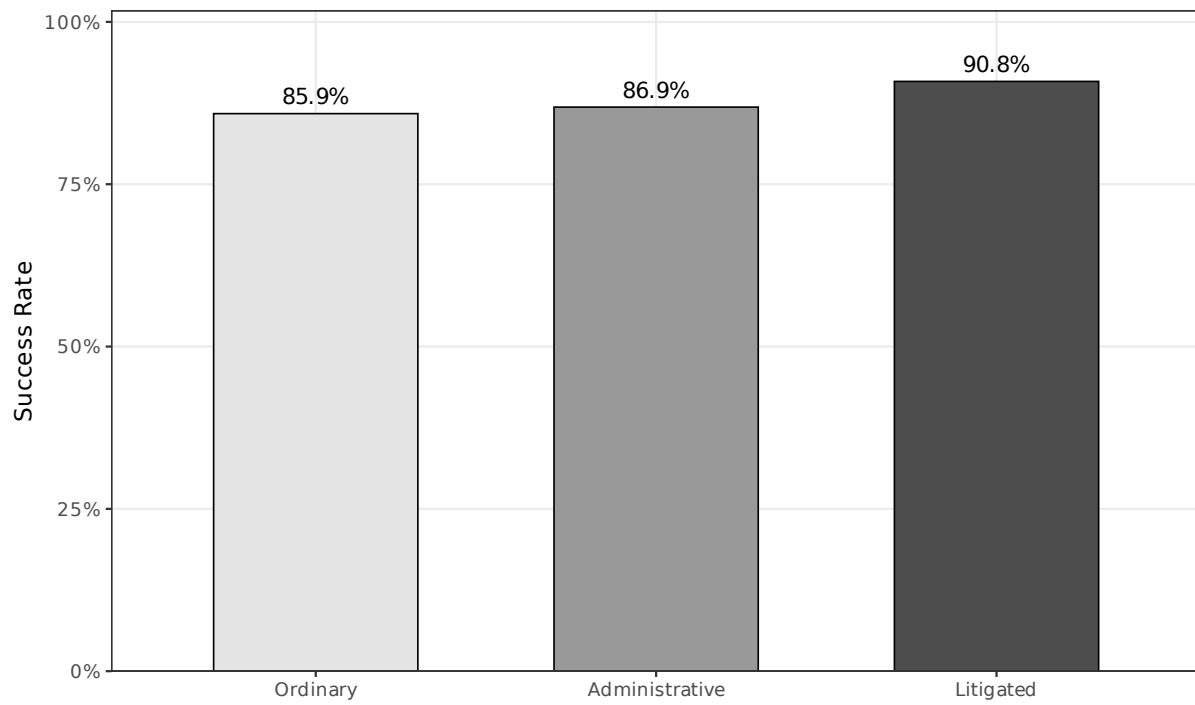


Figure .11: Tender Success Rate by Purchase Type

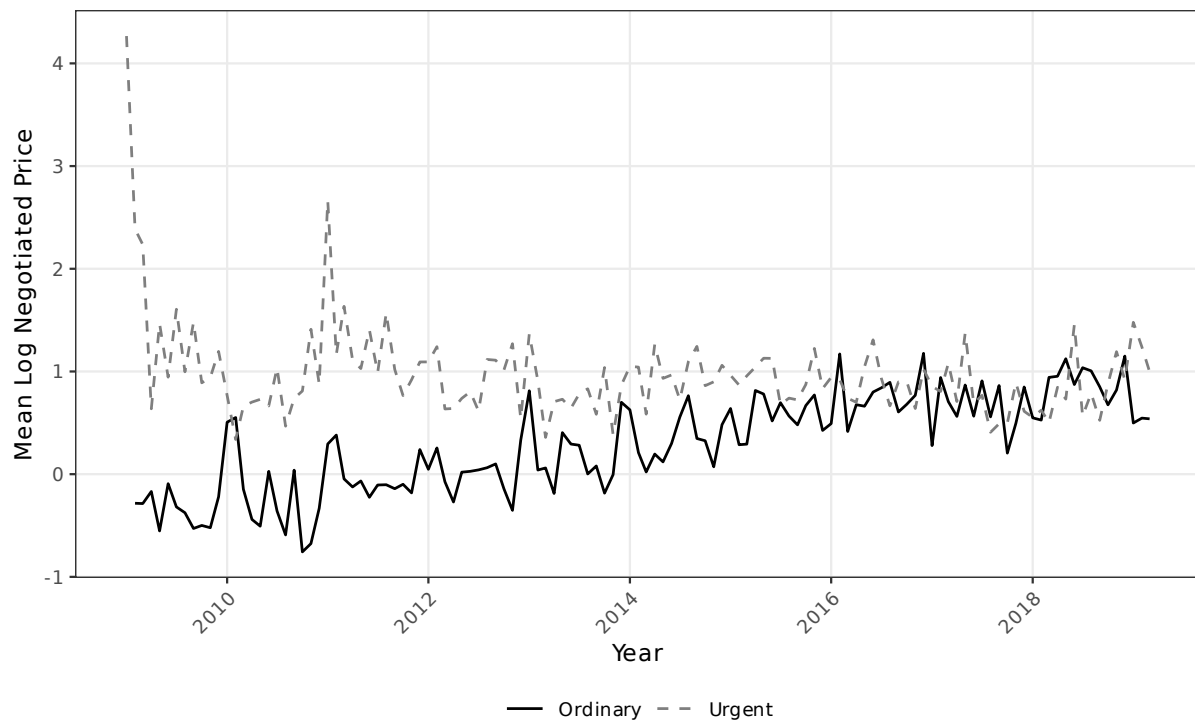


Figure .12: Mean Log Negotiated Price Over Time: Ordinary vs. Urgent

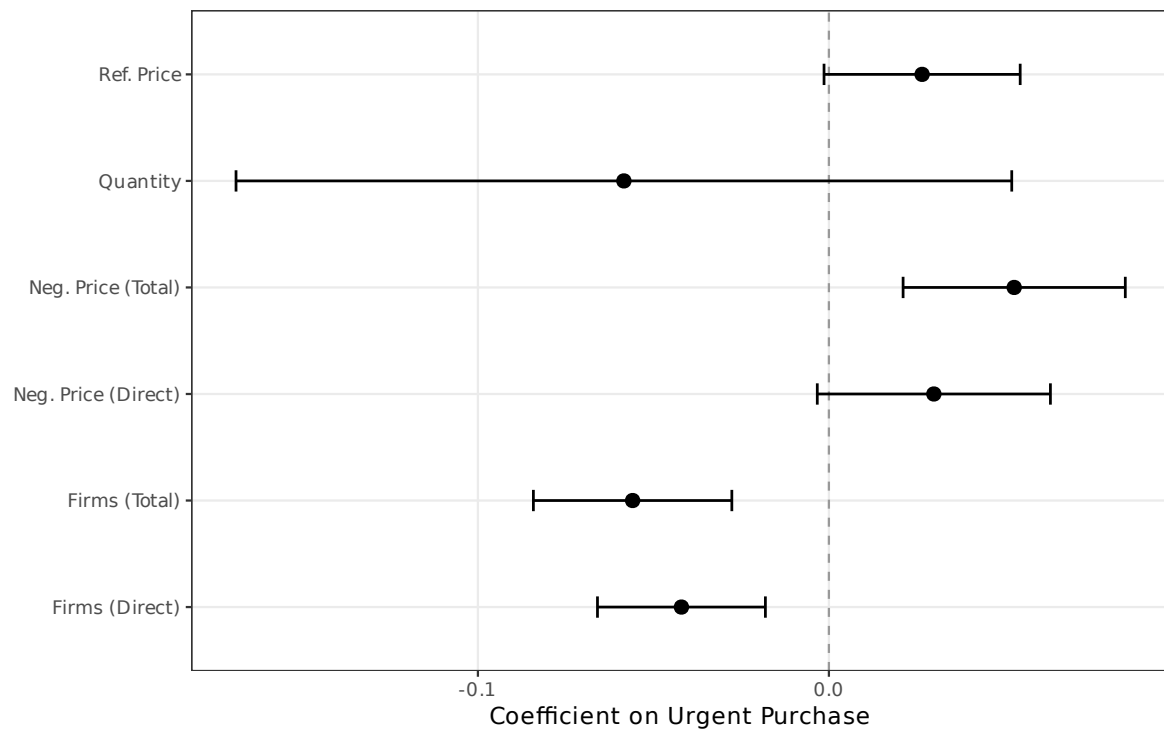


Figure .13: Coefficient Estimates with 95% Confidence Intervals (Preferred Specification)